SIMATS ENGINEERING
ASSESSMENT TOOL 1
COURSE NAME: Artificial Intelligence COURSE CODE: CSA1741

CO1-AT1-Analytical Problem Solving

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks: 50**

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent?
   ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

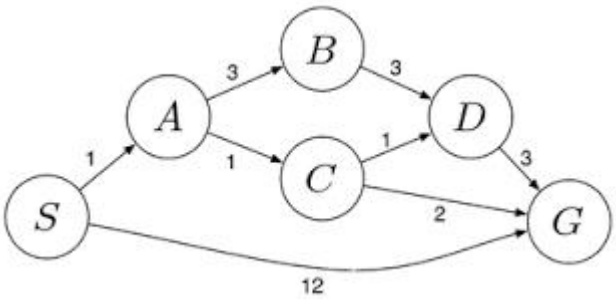
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost

path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



Code of Conduct:

I _____L Swetha_____ (192412678) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: L Swetha

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Problem Understanding	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	10
Method Selection	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	10
Solution Process	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	12.5
Accuracy of Calculation	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	10
Interpretation of Results	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	7.5

ANSWERS

1)A) Problem statement

We are given two jugs:

4-gallon jug (A): Capacity of 4 gallons.

3-gallon jug (B): Capacity of 3 gallons.

Neither jug has measurement markings. We have a water pump and can fill a jug, empty a jug onto the ground, or pour water from one jug into the other.

Goal: Obtain exactly 2 gallons of water in the 4-gallon jug.

Step-by-step solution

Step	Action performed	4-gallon jug (A)	3-gallon jug (B)
1	Initially, both jugs are empty.	0	0
2	Fill the 3-gallon jug completely.	0	3
3	Pour water from B into A.	3	0
4	Fill the 3-gallon jug again.	3	3
5	Pour from B into A until A is full. Only 1 gallon fits in A, leaving 2 gallons in B.	4	2
6	Empty the 4-gallon jug onto the ground.	0	2
7	Pour the remaining 2 gallons from B into A.	2	0

Result: Exactly 2 gallons of water are obtained in the 4-gallon jug by first filling the 3-gallon jug and pouring its water into the 4-gallon jug. Fill the 3-gallon jug again and pour water into the 4-gallon jug until it becomes full, leaving 2 gallons in the 3-gallon jug. Empty the 4-gallon jug and transfer the remaining 2 gallons into it. Thus, the goal is achieved without using any measuring devices.

Explanation

The key step is Step 5. When the 4-gallon jug already contains 3 gallons and the 3-gallon jug is full, only 1 gallon can be transferred into the 4-gallon jug. Therefore, 2 gallons remain in the 3-gallon jug. Emptying the 4-gallon jug and transferring those remaining 2 gallons into it achieves the goal.

Problem formulation in Artificial Intelligence

- Initial state: \((0,0)\), both jugs empty.

- Goal state: $(2,0)$, 2 gallons in the 4-gallon jug and 0 in the 3-gallon jug.
- Actions: Fill, empty, and pour water between jugs.
- State space: All possible water quantities (x,y), where $0 \leq x \leq 4$ and $0 \leq y \leq 3$.
- Search method: State-space search, using Breadth-First Search (BFS) to find a solution with the fewest actions when each action has equal cost.

Conclusion: By filling, pouring, and emptying the jugs in the correct sequence, we obtain exactly 2 gallons in the 4-gallon jug without using any measuring device.

2)A) Mars Rover as an Intelligent Agent

A Mars Rover is an intelligent robotic agent used to explore Mars, collect soil and rock samples, analyse them, and transmit scientific data to Earth.

i. What are the percepts for this agent?

Percepts are the information received through the rover's sensors.

- Camera images: Identify rocks, soil, and surrounding terrain.
- Environmental sensors: Measure temperature, pressure, and radiation.
- Obstacle sensors: Detect rocks, craters, and dangerous slopes.
- Scientific instruments: Analyse the chemical and mineral composition of samples.
- System sensors: Monitor battery level, position, and equipment status.

ii. Characterize the operating environment.

The Mars Rover environment has the following characteristics:

- Partially observable: The rover cannot observe the entire Martian surface.
- Stochastic: Terrain and equipment conditions may produce uncertain outcomes.
- Sequential: Each action affects future decisions.
- Dynamic: Environmental and equipment conditions may change.
- Continuous: Movement, distance, and temperature involve continuous values.

iii. What actions can the agent take?

- Move and navigate across the Martian surface.
- Detect and avoid obstacles.
- Capture images of rocks and soil.
- Collect and analyse samples.
- Transmit scientific data to Earth.
- Monitor battery power and plan exploration routes.

iv. How can one evaluate the performance of the agent?

The rover's performance can be evaluated using the following measures:

- Accuracy of scientific measurements and sample analysis.
- Number and quality of samples collected.

- Distance travelled and area explored.
- Safety and successful obstacle avoidance.
- Energy consumption and time efficiency.
- Successful transmission of data to Earth.

v. What sort of agent architecture is most suitable? Why?

A hybrid agent architecture combining model-based, goal-based, and utility-based agents is most suitable.

- Model-based agent: Maintains information about the terrain and surrounding environment.
- Goal-based agent: Plans actions to achieve goals such as collecting samples and exploring new areas.
- Utility-based agent: Selects actions by balancing scientific value, safety, time, and energy consumption.

Reason: The hybrid architecture helps the rover make intelligent decisions in uncertain conditions, avoid hazards, conserve energy, and achieve its scientific objectives efficiently.

Conclusion

The Mars Rover uses sensors to perceive its environment, performs actions to explore and analyse samples, and evaluates its success through scientific accuracy, safety, and energy efficiency. A hybrid intelligent agent is most suitable because it supports effective planning and decision-making in the challenging environment of Mars.

3)A) Eight-Queens Problem Using Search Strategies

Problem Statement

The 8-Queens problem is a classic Artificial Intelligence search problem. The objective is to place eight queens on an \(8 \times 8\) chessboard such that no two queens attack each other. Therefore, no two queens should share the same row, column, or diagonal.

Problem Formulation

- Initial State: An empty \(8 \times 8\) chessboard.
- State Space: All possible arrangements of eight queens on the chessboard.
- Actions: Place a queen in a safe square, usually one queen per column.
- Transition Model: Place a queen in a selected safe position.
- Goal Test: Check whether all eight queens are placed without attacking one another.
- Path Cost: Each queen placement has a cost of 1, making the total cost of a complete solution

Search Strategy: Backtracking

Backtracking search is commonly used to solve the 8-Queens problem.

- Place a queen in a safe position in the first column.
- Move to the next column and place another queen in a safe row.

- Continue placing queens column by column.
- If no safe position is available, return to the previous column and move its queen to another position.
- Repeat until all eight queens are placed without conflicts.

Example of a Valid Solution

One valid arrangement is represented by the row numbers for columns 1 to 8:

$$[1, 5, 8, 6, 3, 7, 2, 4]$$

This means the queens are placed at row 1 in column 1, row 5 in column 2, row 8 in column 3, and so on. No two queens share a row, column, or diagonal.

Conclusion

The 8-Queens problem can be solved using backtracking search, which explores possible queen positions and rejects invalid configurations. This reduces unnecessary searching and demonstrates how Artificial Intelligence search strategies solve constraint-satisfaction problems.

4)A) OLA Cab Booking Using a Problem-Solving Agent

Suitable Agent

A goal-based problem-solving agent is suitable for the OLA cab booking application. Its goal is to help the customer travel from the pickup location to the destination by selecting a suitable cab based on fare, travel time, availability, and comfort.

Problem Formulation

- Initial State: Customer's current or pickup location.
- Goal State: Customer reaches the destination.
- Actions: Select a cab type, book the cab, and travel to the destination.
- Constraints: Cab availability, fare, travel time, and customer preferences.
- Performance Measure: Reach the destination safely and comfortably at a suitable cost and time.

Pseudocode

```
BEGIN
  Input pickup location and destination
  Input customer preferences
  Find available cabs
  Display cab types: Mini, Micro, Sedan,
                Shared, Prime
  Calculate fare and estimated travel time
  Filter cabs based on preferences
  Select the most suitable cab
  Display cab details and fare

  IF customer confirms booking THEN
```

```
    Book the cab
    Track the journey
    Reach the destination
  ELSE
    Display other available options
  END IF
END
```

Explanation

The agent collects the pickup location, destination, and customer preferences. It identifies available cabs and compares their fares and estimated travel times. Based on the customer's requirements, it recommends a suitable cab. Once the customer confirms, the application books the cab and tracks the journey.

Conclusion

A goal-based problem-solving agent is suitable because it identifies the customer's travel goal, evaluates available cab options, and selects an appropriate option to reach the destination.

5)A) Uniform Cost Search (UCS) Algorithm

Problem Statement

A logistics company wants to transport packages from the starting warehouse S to the destination warehouse G at minimum cost. Each route has a cost representing fuel and travel time. The Uniform Cost Search (UCS) algorithm finds the least-cost path by expanding the node with the lowest cumulative path cost first.

Given Weighted Graph

The routes and their costs are:

- S → A = 1
- S → G = 12
- A → B = 3
- A → C = 1
- B → D = 3
- C → D = 1
- C → G = 2
- D → G = 3

Steps of UCS

Step	Node expanded	Path considered	Cumulative cost
1	S	S → A	1
2	A	S → A → C	2
3	C	S → A → C → D	3
4	D	S → A → C → G	4

5	G	Goal reached	4

UCS always selects the node with the lowest cumulative cost. The direct path S → G has a cost of 12, which is higher than the indirect routes.

Cost Comparison

Path	Total cost
S → G	12
S → A → B → D → G	10
S → A → C → D → G	6
S → A → C → G	4

Final Result

The Uniform Cost Search algorithm finds the least-cost path S → A → C → G, with a total cost of 4 units. This route is optimal because its cumulative cost is lower than that of all other available paths. UCS guarantees an optimal solution when all route costs are non-negative.